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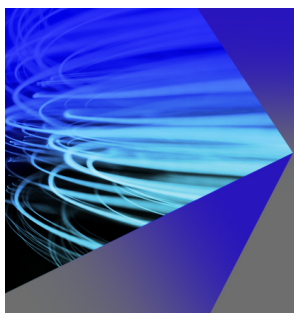
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ABSTRACT

This study investigates the influence of particle roughness on pressure drop in randomly packed beds through experiments and theoretical modeling using the Quinn fluid flow model (QFFM). While classical frameworks like the Ergun and Kozeny–Carman equations assume smooth particles, real-world applications often involve rough or irregularly shaped particles, leading to significant deviations in predicted pressure drops. The QFFM, which integrates Nikuradse’s roughness effects into a modified Ergun equation, is presented as a universal predictor for both packed and empty conduits across all flow regimes (creeping to turbulent). Experiments were conducted on packed beds with spherical and cylindrical particles, with surface roughness quantified via profilometry; additionally, experimental data for irregular particles (clinker) from the literature were used for validation. The QFFM solver accurately back-calculated key parameters (e.g., wall normalization coefficient and porosity) and demonstrated that roughness reduces dimensionless permeability, increasing pressure drop. Results show excellent agreement between measured and predicted data, with mean absolute errors below 1% for smooth particles. A dimensionless analysis reveals distinct scaling for rough vs smooth systems, bridging Nikuradse’s pipe-flow findings to closed conduits. The QFFM’s ability to simultaneously resolve design and operational parameters underscores its utility for industrial applications involving textured or irregular particles.

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I. INTRODUCTION

The study of fluid flow through randomly packed beds of particles is fundamental to numerous industrial and environmental processes, including chemical reactors, filtration systems, and subsurface flows.^{1–3} While the fluid dynamics of smooth spherical particles have been extensively investigated, the effects of surface roughness on flow resistance remain less understood, despite its critical role in practical applications where particle asperities significantly alter local flow structures and energy dissipation.^{4–6}

A cornerstone in the modeling of packed bed flows is the following Ergun equation,⁷ which empirically relates the pressure drop to the superficial velocity through a linear superposition of viscous and inertial losses,

$$\frac{\Delta P}{L} = \underbrace{150 \frac{(1 - \varepsilon)^2}{\varepsilon^3} \frac{\mu u}{d_p^2}}_{\text{Viscous term}} + \underbrace{1.75 \frac{1 - \varepsilon}{\varepsilon^3} \frac{\rho u^2}{d_p}}_{\text{Inertial term}}, \quad (1)$$

where ΔP is the pressure drop [Pa], L is the length of the packed bed [m], ε is the porosity (void fraction) of the bed, μ is the dynamic viscosity of the fluid [Pa s], u is the superficial fluid velocity [m/s], d_p is the particle diameter [m], and ρ is the fluid density [kg/m³].

The Ergun equation has offered a simple empirical framework that combines viscous and inertial flow regimes. However, its formulation assumes idealized smooth spherical particles, leading to significant deviations when applied to rough particles.^{8–10} The key limitations of the Ergun equation include its neglect of surface roughness effects. Its constants (150 for the viscous term and 1.75 for the inertial term) were

derived from smooth-particle experiments and fail to account for microscale turbulence and boundary layer disruptions caused by roughness, which significantly alter the friction factor's dependence on Reynolds number (Re).

An experimental study by Nikuradse¹¹ demonstrates how surface roughness shifts the power-law scaling of ΔP with Re , directly contradicting the Ergun equation's fixed exponents and highlighting its inadequacy for rough particle systems. The dependence of friction factor (f) on Reynolds number in laminar flow, first established by Hagen and Poiseuille over 150 years ago as $f = 64/Re$, was confirmed by Nikuradse to hold for both smooth and rough surfaces in the laminar regime.

Building on this fundamental principle, Kozeny and later Carman developed an analogous relation for porous media, resulting in the Kozeny–Carman equation for pressure drop in packed beds under creeping flow conditions. This equation, derived independently by Kozeny¹² and Carman,¹³ models the packed bed as a network of tortuous capillaries and applies Poiseuille's law for laminar flow through straight pipes to these curved passages, providing a theoretical foundation for predicting pressure drop in low-velocity flows through porous structures,

$$\frac{\Delta P}{L} = 180 \frac{(1 - \varepsilon)^2 \mu u}{\varepsilon^3 d_p^2}. \quad (2)$$

The Kozeny–Carman Eq. (2) is valid for creeping flow regimes where viscous forces dominate, failing to predict pressure drops in transitional or turbulent flows, unlike the more general Ergun equation. The formulation of the Kozeny–Carman equation reveals critical microstructural dependencies, with pressure drop scaling as $(1 - \varepsilon)^2/\varepsilon^3$, making flow resistance highly sensitive to porosity changes and exhibiting inverse quadratic dependence on particle diameter. While theoretically grounded in the Hagen–Poiseuille analogy, the equation assumes hydraulically smooth surfaces.

Numerous experimental studies have validated the Kozeny–Carman equation for low-Reynolds-number flows.^{14–16} However, as the Reynolds number increases, the equation systematically deviates from experimental data for rough particles due to unaccounted-for microscale turbulence and surface-roughness-induced drag.^{17,18} This limitation becomes particularly significant in industrial applications involving textured or irregular particles, where the equation underestimates the observed pressure drop.^{14,19,20}

The Burke–Plummer equation is a semi-empirical model used to predict pressure drop in packed beds at high Reynolds numbers, where inertial forces dominate viscous forces. Unlike the low-Re Kozeny–Carman equation, Burke–Plummer neglects viscous effects, assuming turbulent flow conditions. The equation is expressed as

$$\frac{\Delta P}{L} = \frac{1.75 \rho u^2 (1 - \varepsilon)}{\varepsilon^3 d_p}. \quad (3)$$

In its general form, the Ergun equation combines the Kozeny–Carman and Burke–Plummer equations. However, all three equations express a permeability and internal resistance coefficient as constant, neglecting the influence of particle roughness—a key factor that contradicts Nikuradse's experimental findings for turbulent flow in rough pipes, where a friction factor is a function of Re and relative roughness. Thus, over a wide range of flow velocities, the Ergun equation produces significant errors, potentially leading to incorrect conclusions.^{21,22}

For this reason, many researchers have proposed various modified versions of the equation.

Macdonald²³ published a paper evaluating two friction factor-Reynolds number correlations—the nondimensional Forchheimer equation²⁴ and a modified Ergun equation—using extensive experimental data. Prieur²⁵ developed a theoretical approach to flow resistance by solving the Navier–Stokes equations within pore spaces, introducing a model requiring just one fitting parameter. In a separate study, Niven²⁶ reexamined the Ergun equation through the lens of pore geometry, conceptualizing flow paths as sequences of diverging and converging segments. While this framework produced a comprehensive pressure drop model for granular media, it necessitated the determination of eight additional parameters beyond standard fluid properties and flow velocity.

Additionally, multiple studies have demonstrated the significant influence of particle roughness on pressure drop in packed beds. Crawford *et al.*²⁷ experimentally investigated how surface roughness affects flow resistance in randomly packed beds of spheres, demonstrating that roughness significantly increases pressure drop across both viscous and inertial flow regimes—contrary to conventional pipe-flow analogs. Their study compared smooth spheres with two roughened variants (relative roughness of 0.012 and 0.026), revealing that roughness impacts low-Reynolds-number flows, a finding that challenges prior assumptions. Allen *et al.*²⁸ investigated how particle characteristics (shape, size distribution, packing arrangement, and roughness) affect packed bed pressure drop, demonstrating that the Ergun equation significantly deviates from experimental measurements for non-spherical particles. Their findings reveal that rough spheres, cylinders, cubes, and crushed rock exhibit higher pressure drops than smooth spheres, with particle orientation in irregular packings further influencing flow resistance, necessitating empirical correlations for accurate predictions.

In modified versions of the Ergun equation, particle roughness in the stationary layer is accounted for through the introduction of additional empirical coefficients.^{29–31} Quinn³² introduced the Quinn fluid flow model (QFFM), which integrates fundamental concepts from the Ergun equation with Nikuradse's findings on roughness effects. The derived equation incorporates both particle shape and relative roughness, extending Nikuradse's framework to a packed bed. The QFFM model was validated against experimental data across a broad range of relative roughness values, demonstrating strong agreement between measured and predicted results.^{33,34}

The extensive body of research on flow modeling in the packed beds underscores the ongoing scientific interest in understanding how particle roughness affects pressure drop. While numerical simulations have gained prominence in recent years, their application to packed beds with high particle counts remains constrained by computational limitations. Consequently, developing an accurate model capable of reliably predicting pressure drop across a broad range of operational and geometric parameters remains a critical research objective. This study presents experimental investigations of pressure drop in packed beds filled with rough particles, examining various bed-to-particle diameter ratios. The experimental data were utilized to validate and assess the predictive accuracy of the QFFM in calculating pressure drop.

II. EXPERIMENTS

The experimental setup for investigating flow dynamics in a packed bed is conventional, as illustrated in Fig. 1, which depicts both

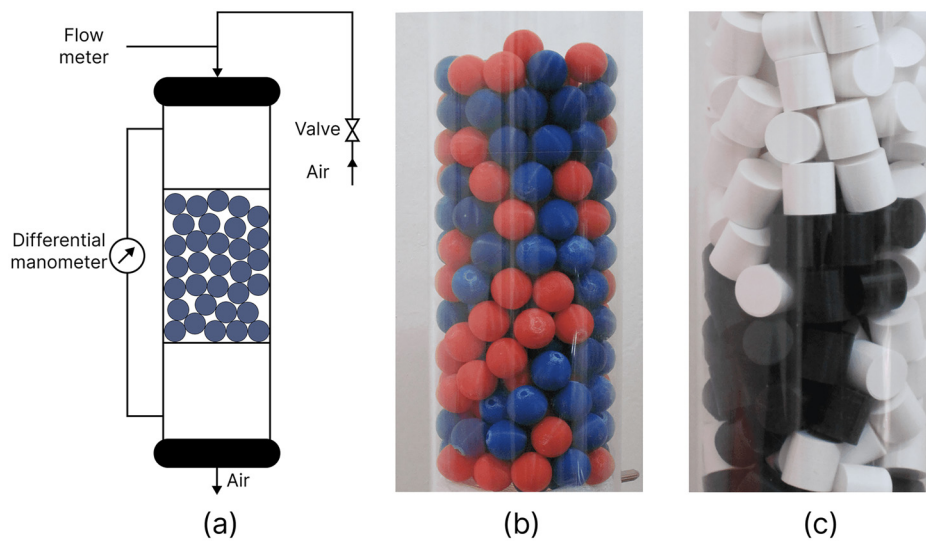


FIG. 1. Schematic representation (a) and photo of packed beds filled with (b) spheres and (c) cylinders.

a schematic diagram and a photograph of the packed beds. The system includes pressure gauge tubes, a flow meter, manometers, and a fan. The inlet air volumetric flow rate is regulated using the flow meter, while the pressure drop across the bed is measured with a differential manometer. Additionally, particle surface roughness is determined using a profilometer (SURFTEST SV-3200, Production Series 178).

The study examined two particle morphologies: cylindrical and spherical. The particles are solid (their porosity is zero). For each morphology, both rough and polished surface variants were investigated. Quantitative surface roughness measurements for all particle variants are presented in Fig. 2.

The flow dynamics through the packed bed were tested at various velocities. To ensure experimental accuracy, each velocity condition was repeated at least twice. The summarized parameters of the packed beds are presented in Table I.

III. QUINN'S FLUID FLOW MODEL

A. QFFM overview

The full derivation of the Quinn fluid flow model is provided in Quinn's previous paper.³⁵ QFFM is based on the following principles:

- (1) The QFFM is focused on pressure-driven fluid flow in closed conduits, which may or may not contain solid particles.
- (2) The model accommodates, seamlessly, both varieties of conduits, meaning it is not relevant whether the conduit contains solid obstacles or is devoid of the same.
- (3) All conduits in the context of the QFFM are considered "packed conduits," which means that the conduits contain a discrete number of particles, n_p , which may or may not contain a solid skeleton. A conduit such as a capillary, for instance, which contains no solid obstacles, is considered a packed conduit containing particles of free space.

For purposes of examining the relationship in a packed bed between the driving force ΔP and the resulting fluid volumetric flow rate, q , the QFFM quantifies a packed bed in terms of its specific anatomy. This is accomplished by defining the packed conduit porosity ε as a

mathematical function of number of particle equivalents of diameter d_p , n_p , and the particle porosity, ε_p , including several subsets of ε , namely:

- Conduit external volume fraction: $\varepsilon_0 = (1 - n_p/n_{pq})$;

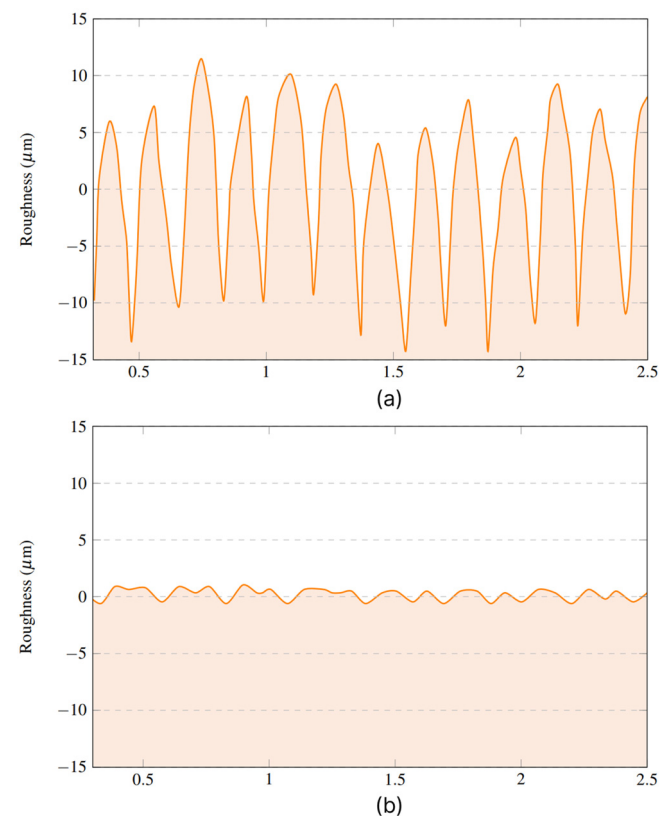


FIG. 2. Particles roughness: (a) cylinders and (b) polished cylinders.

TABLE I. The packed bed parameters.

Parameters	$D/d_p = 4.8$	$D/d_p = 2.9$
Packed bed diameter (D), mm	96	58
Particle size (d_p), mm	20	20
Inlet and outlet diameter, mm	96	58
Fluid	Air	Air
Packed bed length, mm	50–250	50–250
Distance between the inlet and the particle bed starting point, mm	130	65
Distance between the outlet and the particle bed ending point, mm	410	130
Total length of the geometry, mm	1000	600

- Conduit particle volume fraction: $(1 - \varepsilon_0) = n_p/n_{pq}$, where n_{pq} is the number of particle equivalents whose collective volume is equal to the volume of free space within an empty conduit;
- Conduit particle skeletal volume fraction: $\varepsilon_{sk} = (1 - \varepsilon_p)n_p/n_{pq}$;
- Conduit internal volume fraction: $\varepsilon_i = \varepsilon_p(n_p/n_{pq})$;
- Conduit total volume fraction: $\varepsilon_t = 1 - (1 - \varepsilon_p)n_p/n_{pq}$.

Accordingly, having now established a mathematical framework to accommodate a discussion of fluid flow within the packed bed, the QFFM introduces the concept of the Hypothetical Q Channel (HQC) in terms of its characteristic dimension, i.e., its diameter $d_c = d_p/\text{abs}(1 - \varepsilon_0)$, which is the ratio of the spherical particle diameter equivalent and the magnitude of the particle volume fraction.

The QFFM is based upon a concept of fluid flow that is more than 100 years old.³⁶ The underlying theory is systematically derived from that concept, consisting of a universal relationship between fluid flow rate, q , and differential pressure, ΔP , using 10 sequential development steps. These steps are as follows:

1. Pressure drop in a general form:

$$\frac{\Delta P}{L} = a_{\text{For}}q + b_{\text{For}}q^2, \quad (4)$$

where a_{For} , b_{For} are Forchheimer coefficients that are empirical coefficients to balance the equation.

2. Forchheimer equation:

$$\frac{\Delta P}{\rho_f g L} = a_{\text{For}}\mu_s + b_{\text{For}}\mu_s^2, \quad (5)$$

where $\frac{\Delta P}{\rho_f g L}$ is the hydraulic gradient; ρ_f is the fluid density; g is the acceleration due to gravity; and μ_s is the fluid superficial velocity.

3. Substitute for a_{For} and b_{For} :

$$\frac{\Delta P}{\rho_f g L} = \frac{4r_h^3\pi\delta\eta\mu_s}{3\rho_f g d_c^2} + \frac{\delta^2\lambda\mu_s^2}{2\pi g d_c}, \quad (6)$$

where $r_h = 4$ is the ratio of the surface area to cross-sectional area of a spherical particle; $\delta = 1/\varepsilon_0^3$; η is the fluid absolute viscosity; and λ is the wall normalization coefficient.

4. Rearrange by multiplying across by $\rho_f g$:

$$\frac{\Delta P}{L} = \frac{268\delta\eta\mu_s}{d_c^2} + \frac{\delta^2\lambda\rho_f\mu_s^2}{2\pi d_c}. \quad (7)$$

5. Judicious grouping of terms:

$$\frac{\Delta P}{L} = 268n_v + \frac{\delta\lambda n_k}{2\pi}, \quad (8)$$

where n_v , n_k are the viscous hydraulic force and kinetic hydraulic force exerted per unit element of fluid control volume, respectively; $n_v = \frac{\delta\eta\mu_s}{d_c^2}$; $n_k = \frac{\delta\rho_f\mu_s^2}{d_c}$.

6. Normalize for viscous terms by dividing across by n_v :

$$\frac{\Delta P}{n_v L} = 268 + \frac{\delta\lambda n_k}{2\pi n_v}. \quad (9)$$

7. Identify the Q-modified Ergun equation:

$$f_v = A + B\text{Re}_m, \quad (10)$$

where $f_v = \frac{\Delta P}{n_v L}$ is the viscous friction factor; $A = 268$ is the viscous constant; $B = \frac{\delta\lambda}{2\pi}$ is the kinetic constant; $\text{Re}_m = \frac{n_k}{n_v}$ is the modified Reynolds number. It should be noted that the fluid motion term is the product of two entities: n_v , the viscous hydraulic force exerted per unit element of fluid control volume and $(4\pi r_h^3/3) = 268$ also known as the “viscous constant,” where r_h is the ratio of surface area to cross-sectional area of a spherical particle, i.e., $r_h = 4$. Incidentally, we point out that the value of 270 for the viscous constant in the Kozeny/Blake equation was identified by Giddings.³⁷

8. Normalize for r_h (hydrodynamic drag) by dividing across by $r_h = 4$:

$$\frac{f_v}{4} = 67 + \frac{\delta\lambda\text{Re}_m}{8\pi}. \quad (11)$$

9. Identify the universal flow equation (Quinn’s Law):

$$P_Q = k_1 + k_2 C_Q, \quad (12)$$

where $P_Q = \frac{f_v}{r_h}$ is the fluid-drag normalized dimensionless pressure gradient, for spherical particles $P_Q = \frac{f_v}{4}$; $k_1 = \frac{64\pi}{3} \approx 67$ —universal viscous constant; $k_2 = \frac{1}{8\pi} \approx 0.04$ —universal kinetic constant; $C_Q = \delta\lambda\text{Re}_m$.

10. Rearrange terms to identify the impact of wall normalization coefficient (λ) via the dimensionless permeability (Θ) of the HQC:

$$\Theta = \frac{Q_N}{P_Q} = \frac{1}{\frac{k_1}{Q_N} + \frac{\lambda}{2\pi r_h}}, \quad (13)$$

where $Q_N = \delta\text{Re}_m$ is the instantaneous fluid current.

Based on the mathematical transformations described in detail in the paper,³⁵ we obtain an expression for determining the pressure drop in the packed bed,

$$\frac{\Delta P}{L} = \frac{A(1 - \varepsilon_0)^2\mu_s\eta}{\varepsilon_0^3 d_p^2} + \frac{B(1 - \varepsilon_0)\rho_f\mu_s^2}{\varepsilon_0^3 d_p}. \quad (14)$$

We point out that Eq. (14) corresponds to the general format of the Ergun equation. However, we call it the Q-modified Ergun

equation because the “constants” of $A = 268$ (approximately) and $B = \lambda/(2\pi\epsilon_0^3)$ have been modified from Ergun’s original constant values of 150 and 1.75 for A and B , respectively, and most importantly, B is not a constant, but rather is a function of both λ and ϵ_0 .

B. Wall normalization coefficient

In Eq. (6), wall normalization coefficient, λ , was introduced. The wall normalization coefficient, λ , is a dimensionless parameter that quantifies the total additional flow resistance (or “wall effect”) arising from the presence of solid boundaries.

Its physical meaning can be broken down as follows:

- $\lambda = 1$: This is the baseline value for a theoretically ideal packed bed with no wall effects. This signifies that the energy loss is purely due to the frictional and inertial forces predicted for the core flow through the porous structure itself, with no additional resistance from viscous boundary layers or surface roughness.
- $\lambda > 1$: A value greater than 1 indicates a net increase in flow resistance due to one or both of the following “wall effects”:
 - Primary effect (viscous boundary layer): An increase in λ caused by the dissipation of energy within a viscous boundary layer that develops along the confining walls of the conduit or the surfaces of particles. This effect is often dominant in smooth-walled systems with low tortuosity (e.g., empty pipes, large D/d_p beds).
 - Secondary effect (surface roughness): An increase in λ caused by energy losses due to flow separation, recirculation, and enhanced turbulence triggered by surface roughness on the particles or the conduit walls.

C. QFFM solver

Because the QFFM includes all variables involved in the pressure/flow relationship in closed conduits, it can be used to evaluate the results of experimental data with a view to enabling the prediction of the performance of a given packed conduit under study, over the entire region of the fluid flow regime, from creeping to fully turbulent flow. This is accomplished by what we refer to as the use of the QFFM solver. Therefore, for instance, difficult to measure variables such as, d_p , ϵ_0 , and k are the ones of most interest in this scenario. Furthermore, because the Conservation Laws dictate that the variables of d_p and ϵ_0 are dependent variables in the pressure/flow relationship, the QFFM solver delivers the necessary simultaneous solution for both. This simultaneous solution for the variables of d_c and δ is defined as follows:

$$d_c = \frac{1}{X^{(1/6)}Y^{(1/2)}}, \quad (15)$$

$$\delta = \frac{X^{(1/6)}}{Y^{(1/2)}}, \quad (16)$$

where $X = \alpha\beta$ and $Y = \frac{\alpha}{\beta}$ and $\alpha = \frac{\delta}{d_c^2} = \frac{3a\rho_f g}{4r_h^3 \pi \eta}$ and $\beta = \frac{\delta^2}{d_c} = \frac{2b\pi g}{\lambda}$.

As can be seen from the definition of α and β , in experimental results, all variables are known except the variables of a , b , and λ . The values of a and b are the Forchheimer coefficients (a_{For} and b_{For}) and are determined from the experimental results expressed in the form of the hydraulic gradient as explained earlier in Eq. (5). However, in order

to obtain accurate values for both Forchheimer coefficients, i.e., the measured and calculated values for pressure drop must be balanced, all variables including the difficult to measure ones, d_p , ϵ_0 , k , and λ must be determined in advance.

D. Methodology of applying the QFFM solver to experimental data

The methodology of applying the QFFM solver to permeability experimental data when the sand-roughness coefficient is unknown involves a number of sequential steps outlined below. These steps apply to either an empty conduit or a conduit containing obstacles. It should be pointed out, however, that at least one of the variables of d_p or ϵ_0 has to be measured and provided as a given in order to isolate the value of k , the sand-roughness coefficient.

- Assume that the value of k , the sand-roughness coefficient, is zero in the QFFM permeability equation, i.e., smooth particles.
- Determine the Forchheimer coefficients a and b using the QFFM solver to deliver the correct value for the given variable, i.e., d_p or ϵ_0 . This solution will independently identify the value of λ due to the primary wall effect, if any, for smooth particles.
- Set this value of λ in the QFFM and use the solver again to reconcile the given variable, d_p or ϵ_0 , by updating the value of δ .
- If the measured pressure drop is still larger than the QFFM calculated values in the previous step, then the particles are rough.
- Balance the calculated pressure drop with the larger measured pressure drop by setting the value of k in the QFFM equation. This solution will independently identify the value of λ due to the secondary wall effect for rough particles.
- Set this value of λ in the QFFM and use the solver again to reconcile the given variable, d_p or ϵ_0 , by updating the value of δ .
- The measured and calculated values for pressure drop should now be balanced. This is the final solution.

IV. RESULTS AND DISCUSSION

We present multiple experimental datasets to evaluate the effect of particle roughness on pressure drop, while simultaneously demonstrating the capabilities of the QFFM solver and validating its predictions:

- Set 1_{sp}: packed bed (4.8 cm diameter) with polished spherical particles;
- Set 2_{sp}: packed bed (9.6 cm diameter) with polished spherical particles;
- Set 1_{cr}: packed bed (4.8 cm diameter) with rough cylindrical particles;
- Set 2_{cp}: packed bed (4.8 cm diameter) with polished cylindrical particles;
- Set 3_{cr}: packed bed (9.6 cm diameter) with rough cylindrical particles.

Additionally, we incorporated experimental results from Buchwald’s paper²⁹ for irregularly shaped particles. This comprehensive approach enables assessment of particle roughness effects on pressure drop across a broad spectrum of operational and design parameters, while providing extensive validation of the QFFM solver.

TABLE II. Design parameters of packed beds filled with polished spherical particles and their corresponding QFFM-calculated parameters.^a

Dataset	Cond.		Part. Diam. d_p (cm)	Ratio $\frac{D}{d_p}$	Rough. Coeff. k (cm)	Wall Coeff. λ	Part. Num. n_p	Ext. Por. ϵ_0	HQC d_c (cm)	Por. Coeff. δ	Tort. τ	Ergun	
	D (cm)	L (cm)										A $\frac{64\pi}{3}$	B $\frac{\delta\lambda}{2\pi}$
Set 1s _p	4.8	25	2.00	2.90	0	1.002 87	176	0.5928	4.91	4.80	176	268.19	0.77
	4.8	20	2.00	2.90	0	1.002 63	144	0.5831	4.80	5.04	185	268.19	0.80
	4.8	15	2.00	2.90	0	1.002 63	108	0.5820	4.78	5.07	186	268.19	0.81
	4.8	10	2.00	2.90	0	1.002 33	75	0.5658	4.61	5.52	202	268.19	0.88
	4.8	5	2.00	2.90	0	1.002 10	39	0.5514	4.46	5.96	218	268.19	0.95
Set 2s _p	9.6	5	2.00	4.80	0	1.000 25	187	0.4593	3.70	10.32	1713	268.19	1.64
	9.6	10	2.00	4.80	0	1.000 26	371	0.4636	3.73	10.04	1665	268.19	1.60
	9.6	15	2.00	4.80	0	1.000 22	569	0.4509	3.64	10.91	1809	268.19	1.74
	9.6	20	2.00	4.80	0	1.000 24	760	0.4502	3.64	10.96	1818	268.19	1.74
	9.6	25	2.00	4.80	0	1.000 22	964	0.4420	3.58	11.58	1921	268.19	1.84

^aParameters k , λ , ϵ_0 , d_c , δ , and τ are outputs calculated by the QFFM solver.

A. Spherical particles

Design parameters of packed beds for Set 1s_p and Set 2s_p as well as their corresponding derivatives for the QFFM solver implementation are presented in Table II. As shown in Table II, we have applied the QFFM solver to these datasets and determined that the spherical particles are hydrodynamically smooth having a sand roughness coefficient $k = 0$. Thus, there is no secondary wall effect in these datasets. There is, however, a small primary wall effect due to the viscous boundary layer as reflected in the value for $\lambda = 1.003$ approximately. Note that the 4.8 diameter packed beds have a larger value for external porosity ϵ_0 than the 9.6 diameter conduits. Table II includes all the relevant QFFM calculations underlying the analysis.

There is no ambiguity in the value of d_p for these spherical particles, and this measured value is therefore given at a value of 2.0 cm in our application of the QFFM solver. Accordingly, the QFFM solver returns an accurate value for external porosity ϵ_0 for these conduits. Consequently, the analysis clearly demonstrates that these particles are indeed hydrodynamically smooth, i.e., the measured and calculated pressure drops are balanced (according to the methodology in Sec. III D). To illustrate a good correlation between experimental and calculated results, we show a correlation example for smooth particles ($k = 0$) where we compare the measured values to the QFFM calculated values as depicted in Fig. 3. A mean absolute error (MAE) was used to elaborate on the correlation between measured and calculated results,

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (17)$$

Here, y_i and $\hat{y}_i$ are predicted (simulated) and expected (experimental) data points. This metric was preferred over the popular root mean squared error ($RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$) because MAE will not exaggerate large errors. That is critical in the considered case, as errors are not normalized. The MAE analysis revealed that MAE for the given design and operational parameter is about 1%. This confirms the

legitimacy of the QFFM solver methodology for smooth spherical particles.

One of the main advantages of the QFFM is its ability to express pressure drop in dimensionless form as a function of both design and operational parameters. Equation (13) defines the dimensionless permeability (Θ) as a function of the instantaneous fluid current (Q_N) for given design parameters, while incorporating particle roughness effects through the wall normalization coefficient (λ). Consequently, the dimensionless parameter Θ in the QFFM effectively captures the influence of particle roughness. The agreement between measured and predicted values is demonstrated in Fig. 4.

Figure 4 includes both the theoretical line for $\lambda = 1$ (representing smooth particles) and experimental data for Set 1s_p and Set 2s_p. The

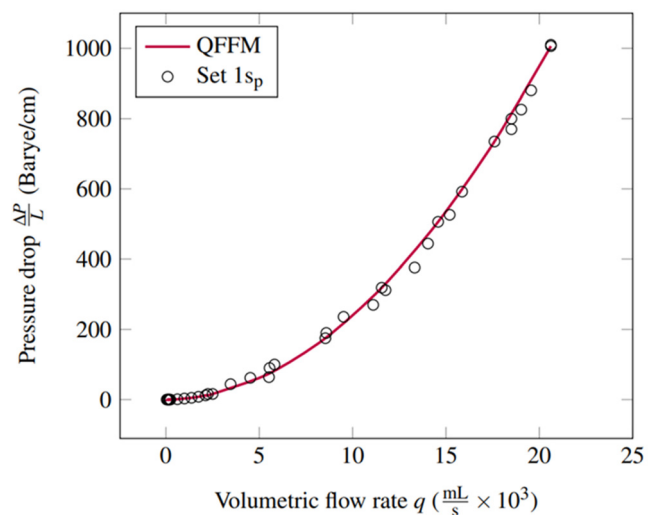


FIG. 3. Pressure drop as a function of volumetric flow rate: line—predicted data via QFFM, points—measured data for Set 1s_p.

results demonstrate excellent agreement between measured data and QFFM predictions for polished spherical particles when the wall normalization coefficient corresponds to a smooth channel ($\lambda = 1$).

All experimental datasets cluster near the $\lambda = 1$ line, confirming that the particles exhibit effectively smooth surface characteristics. This observation validates the QFFM's ability to accurately predict flow behavior for smooth spherical particles in packed beds.

B. Cylindrical particles

The experimental data for packed beds containing rough and polished cylindrical particles were compared with predictions from the QFFM solver. The design and operational parameters of the packed beds were used to calculate the derivatives required for the QFFM solver. We applied the QFFM solver to this dataset, and the calculated results are presented in Table III. The packed beds were tested for three different configurations, as described previously (Set 1_c, Set 2_c, Set 3_c).

Table III demonstrates the significant effect of particle roughness on the wall normalization coefficient (λ). For polished particles, this coefficient is approximately unity ($\lambda \approx 1$), whereas for rough particles, it increases to $\lambda \approx 1.1$. This difference directly influences the coefficient B in the Q-modified Ergun Eq. (10). Consequently, the experimental data and QFFM predictions provide clear evidence of how particle roughness affects the pressure drop in packed beds.

The experimental data and calculated results were expressed in dimensionless form to examine the relationship between dimensionless permeability and instantaneous fluid current, as presented in Fig. 5. Figure 5 demonstrates excellent agreement between experimental and modeled data for polished cylindrical particles. This behavior mirrors the results observed for polished spherical particles in Fig. 4, confirming the consistency of the QFFM approach for smooth particles.

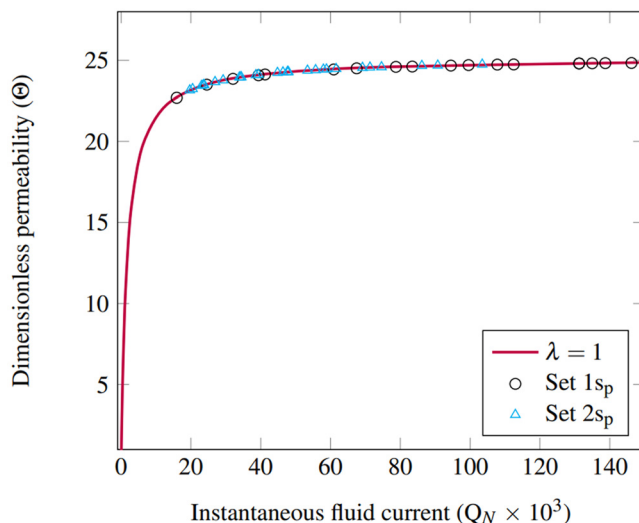


FIG. 4. Dimensionless permeability (Θ) as a function of instantaneous fluid current (Q_N) for a packed bed of spherical particles: line—predicted data via QFFM, points—measured data for Set 1_{sp} and Set 2_{sp}.

Figure 5 further reveals the influence of particle roughness on dimensionless permeability, as evidenced by two distinct rough particle datasets. The results clearly show that surface roughness reduces dimensionless permeability. Since permeability is inversely proportional to pressure drop, these findings demonstrate that particle roughness significantly increases pressure drop in packed beds.

The application of the QFFM model for predicting pressure drop in packed beds containing rough particles offers a universal approach applicable across a wide range of design and operational parameters. In its general form, Eq. (14) maintains the structure of the classical Ergun equation through parameters A and B . However, unlike the original Ergun equation where B remains constant, the Q-modified Ergun Eq. (14) treats B as a function of both the wall normalization coefficient (λ) and the initial porosity (ϵ_0).

C. Irregularly shaped particles

Packed beds containing irregularly shaped particles represent a particularly relevant practical case. To evaluate the effect of particle roughness on pressure drop in such systems, we compared experimental measurements with QFFM model predictions. The experimental dataset, obtained from Buchwald's paper,²⁹ consists of irregularly shaped concrete particles that underwent thermal treatment (denoted by the authors as clinker i^{th}).

Using Buchwald's experimental data, we derived the necessary parameters for the QFFM model, as summarized in Table IV.

Figure 6 presents the dimensionless permeability (Θ) as a function of instantaneous fluid current (Q_N) for a packed bed of irregularly shaped particles. The plot also includes the corresponding relationship for hypothetical smooth particles in an equivalent packed bed configuration.

The results demonstrate that the clinker particles exhibit significant surface roughness, with wall normalization coefficients (λ) exceeding 3. Consequently, the permeability of these rough particle beds is substantially lower than that of hypothetical smooth-particle beds. This dimensionless representation of permeability vs fluid current enables systematic evaluation of packed bed performance across various operational and design parameters.

D. Summary of comparison

The QFFM is a universally accurate predictor of permeability in packed conduits regardless of whether the packed conduits contain solid obstacles or not. Accordingly, it may be used, advantageously, to design any form of packed conduit by simply inputting the known physical characteristics of the conduit, particles, and fluid. If, however, some of the characteristics of the systems are not known, then the QFFM is best utilized by using it in its "solver" capacity. The QFFM solver has the ability to take some basic information, i.e., known values for D , L , q , ΔP , η , ρ_f and back-calculate for all remaining information necessary to do a complete characterization of any packed conduit under study. It has this capability because the solver contains the simultaneous solution for the characteristics of both solid matter and free space occupied by any fluid in any closed conduit under study. Furthermore, it delivers accurate results regardless of where in the fluid flow regime a study happens to fall, i.e., creeping to fully turbulent flow. This differentiates it from all extant fluid flow models.

TABLE III. Design parameters of packed beds filled with rough and polished cylindrical particles and their corresponding QFFM-calculated parameters.^a

												Ergun		
Particle type	Cond.		Part. size		Ratio	Rough. Coeff.	Wall Coeff.	Part. Num.	Ext. Por.	HQC	Por. Coeff.	Tort.	A	B
	D (cm)	L (cm)	d_p (cm)	$\frac{D}{d_p}$	k (cm)	λ	n_p	ε_0	d_c (cm)	δ	τ	$\frac{64\pi}{3}$	$\frac{\delta\lambda}{2\pi}$	
Set 1c _r	4.8	5	2.00	4.64	8.7977×10^{-8}	1.103 65	33	0.4178	2.15	13.71	2055	268.19	2.41	
	4.8	10	2.00	4.64	8.7977×10^{-8}	1.105 89	71	0.3782	2.01	18.49	2771	268.19	3.25	
	4.8	15	2.00	4.64	8.7977×10^{-8}	1.103 89	101	0.4142	2.13	14.07	2109	268.19	2.47	
	4.8	20	2.00	4.64	8.7977×10^{-8}	1.103 13	132	0.4276	2.18	12.79	1916	268.19	2.24	
	4.8	25	2.00	4.64	8.7977×10^{-8}	1.102 36	161	0.4403	2.23	11.72	1756	268.19	2.06	
Set 2c _p	4.8	5	2.00	4.64	0	1.000 44	28	0.5179	2.59	7.20	1079	268.19	1.15	
	4.8	10	2.00	4.64	0	1.000 60	53	0.5405	2.72	6.33	949	268.19	1.01	
	4.8	15	2.00	4.64	0	1.000 63	76	0.5630	2.86	5.60	840	268.19	0.89	
	4.8	20	2.00	4.64	0	1.000 65	101	0.5617	2.85	5.64	846	268.19	0.90	
	4.8	25	2.00	4.64	0	1.000 72	122	0.5757	2.95	5.24	785	268.19	0.83	
Set 3c _r	9.6	5	2.00	7.68	8.7977×10^{-8}	1.104 21	137	0.4062	2.10	14.92	10 140	268.19	2.62	
	9.6	10	2.00	7.68	8.7977×10^{-8}	1.101 14	250	0.4576	2.30	10.44	7093	268.19	1.83	
	9.6	15	2.00	7.68	8.7977×10^{-8}	1.100 83	371	0.4628	2.33	10.09	6855	268.19	1.77	
	9.6	20	2.00	7.68	8.7977×10^{-8}	1.101 24	501	0.4562	2.30	10.53	7156	268.19	1.84	
	9.6	25	2.00	7.68	8.7977×10^{-8}	1.101 45	630	0.4530	2.29	10.76	7309	268.19	1.88	
	9.6	30	2.00	7.68	8.7977×10^{-8}	1.101 16	751	0.4566	2.30	10.50	7138	268.19	1.84	

^aParameters k , λ , ε_0 , d_c , δ , and τ are outputs calculated by the QFFM solver.

In Fig. 7, we summarize the obtained results in this paper. Moreover, because the QFFM can be utilized to predict pressure drop in packed conduits regardless of whether they contain solid particles, we include in this summary the fundamental results obtained by Nikuradse¹¹ for empty conduits with both smooth and rough walls.

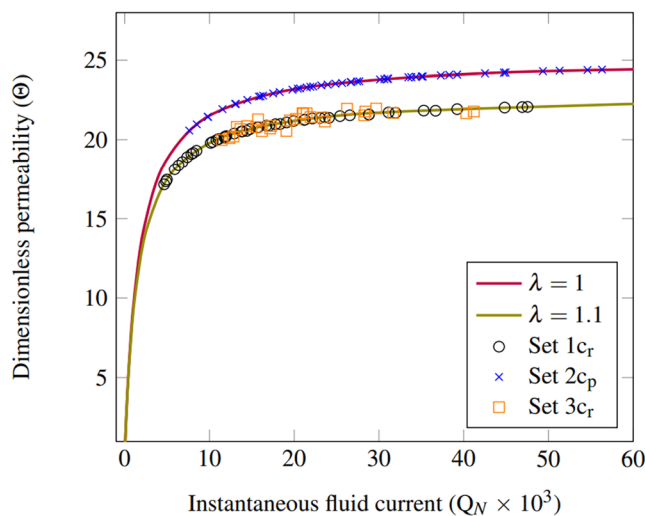


FIG. 5. Dimensionless permeability (Θ) as a function of instantaneous fluid current (Q_N) for a packed bed of cylindrical particles: line—predicted data via QFFM, points—measured data for Set 1_{C_r}, Set 2_{C_p}, Set 3_{C_r}.

Figure 7 shows dimensionless permeability (Θ) as a function of instantaneous fluid current (Q_N) for various systems, demonstrating the unification of pipe and packed bed flow models. The QFFM ($\lambda = 1$) curve for a packed bed of smooth spheres shows a linear relationship, characteristic of a unified model that seamlessly bridges flow regimes. The smooth wall curve for an empty pipe shows a distinct inflection point marking the transition from laminar ($Re < \sim 2100$) to turbulent ($Re > 4000$) flow, where permeability decreases due to turbulent energy dissipation. Data for rough cylindrical particles ($\lambda \approx 1.1$) and clinker ($\lambda \gg 1$) show how increased effective flow resistance further reduces permeability. The data for rough pipes from Nikuradse¹¹ is included for comparison.

The data in Fig. 7 is interpreted as follows:

- The line representing $\lambda = 1$ corresponds to that of a conduit packed with solid particles in which there are no wall effects, either primary or secondary. There is no primary wall effect because the ratio of D/d_p is large, meaning that the tortuosity is also large and, accordingly, there is virtually no viscous boundary layer. There is no secondary wall effect because the particles are smooth. The packed bed filled with spherical particles fits this category and, therefore, the measured data fall on this line. The cylindrical particles have a roughness leading to a visible value for λ . Therefore, dimensionless permeability for the packed bed of rough cylinders is lower than that for the packed bed of polished spheres.
- The line representing smooth walls corresponds to a conduit packed with particles of free space, i.e., an empty conduit, which has smooth walls. This line is different from the line for $\lambda = 1$ because this category has a very significant primary wall effect,

TABLE IV. Design parameters of packed beds filled with irregularly shaped particles and their corresponding QFFM-calculated parameters.^a

Particle type	Cond.											Ergun	
		Part. Size		Ratio	Rough. Coeff.	Wall Coeff.	Part. Num.	Ext. Por.	HQC	Por. Coeff.	Tort.	A	B
		D (cm)	L (cm)	d_p (cm)	$\frac{D}{d_p}$	k (cm)	λ	n_p	ε_0	d_c (cm)	δ	τ	$\frac{64\pi}{3}$
Clinker 1	19.40	15.70	0.104	187	$1.202\,96 \times 10^{-3}$	5.98	3 144 875	0.6049	0.2623	4.52	44 435 035	268.19	4.30
Clinker 1.25	19.40	9.70	0.107	182	$1.229\,64 \times 10^{-4}$	3.29	1 756 479	0.6124	0.2748	4.35	39 458 732	268.19	2.28
Clinker 1.6	19.40	27.10	0.139	139	$5.143\,64 \times 10^{-4}$	4.40	2 240 972	0.6049	0.3523	4.52	18 342 710	268.19	3.16
Clinker 2	19.40	30.00	0.193	100	$3.424\,83 \times 10^{-4}$	3.72	983 586	0.5822	0.4620	5.07	7 715 386	268.19	3.00
Clinker 2.5	19.40	49.50	0.258	75	$8.760\,97 \times 10^{-4}$	4.37	682 934	0.5826	0.6170	5.06	3 242 558	268.19	3.52
Clinker 3.15	19.40	49.60	0.272	71	$1.246\,89 \times 10^{-4}$	2.74	589 966	0.5768	0.6423	5.21	2 841 177	268.19	2.27
Clinker 4	19.40	19.10	0.357	54	$3.224\,48 \times 10^{-4}$	3.17	99 917	0.5790	0.8476	5.15	1 241 751	268.19	2.60
Clinker 5	19.40	36.10	0.378	51	$2.251\,10 \times 10^{-4}$	2.88	156 086	0.5851	0.9119	4.99	1 009 429	268.19	2.29
Clinker 6.3	19.40	44.20	0.479	41	$3.753\,00 \times 10^{-4}$	3.07	94 735	0.5835	1.1494	5.03	502 468	268.19	2.45
Clinker 8	19.40	49.30	0.633	31	$2.896\,55 \times 10^{-4}$	2.73	46 390	0.5775	1.4980	5.19	224 324	268.19	2.26

^aParameters k , λ , ε_0 , d_c , δ , and τ are outputs calculated by the QFFM solver.

i.e., it has a thick viscous boundary layer. The thick viscous boundary layer is due to the fact that the tortuosity of an empty conduit is very small and this leads to a large value of λ . Note, especially, that this line then acts as the differentiating line between a roughened wall of an empty conduit (Nikuradse rough) and that of a conduit packed with roughened solid particles (clinker).

- The clinker data fall on the left-hand side of the smooth wall line because this packed conduit has no viscous boundary layer to obscure the impact of roughness at any value of Q_N , large or small.
- The Nikuradse rough data fall on the right-hand side of the smooth wall line because it has a significant viscous boundary layer, which obscures the impact of the roughness at low values of the Q_N number. The impact of the roughness is only apparent when the roughness punches through the viscous boundary layer

at high values of the Q_N number as the boundary layer increasingly dissipates.

In summary, Fig. 7 clearly differentiates data based upon wall effect (λ) whether that wall effect is primarily due to viscous boundary layer or secondarily due to either solid particle roughness or inner conduit wall roughness.

The QFFM provides a continuous representation from a packed bed to an empty pipe by considering the limiting values of its parameters. The Nikuradse data for flow in empty pipes correspond to the following specific limits within the QFFM framework:

- Particle porosity (ε_p): $\varepsilon_p \rightarrow 1$. This indicates the “particles” are entirely void space (i.e., no solid obstacles exist in the conduit).
- Conduit total porosity (ε_t): $\varepsilon_t = 1$. This signifies that the entire volume of the conduit is available to the fluid and, therefore, for flow, which is true for an empty pipe.

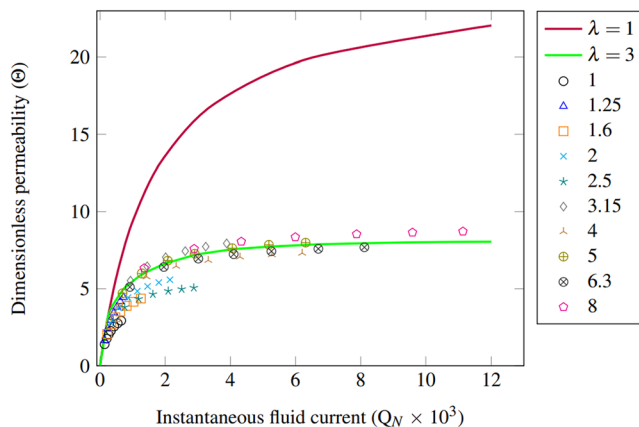


FIG. 6. Dimensionless permeability (Θ) as a function of instantaneous fluid current (Q_N) for a packed bed of irregularly shaped particles: line—predicted data via QFFM, points—measured data from Ref. 29.

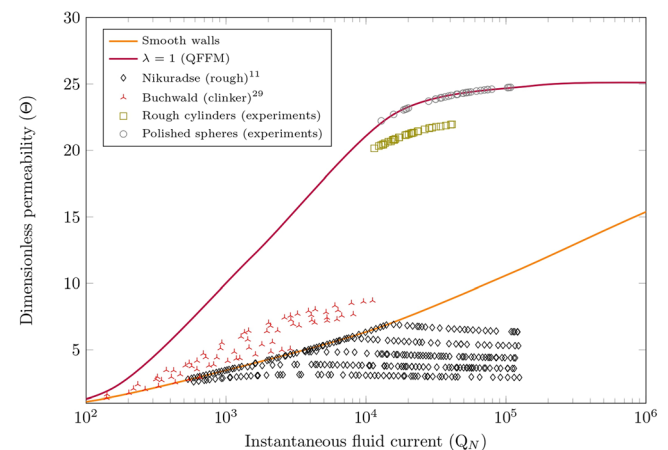


FIG. 7. Dimensionless permeability (Θ) as a function of instantaneous fluid current (Q_N) for various systems.

- Characteristic conduit diameter (d_c): $d_c \rightarrow D$, where D is the actual pipe diameter. In the absence of any solid packing, the effective hydraulic diameter of the flow path becomes the pipe diameter itself.
- Wall normalization coefficient (λ): $\lambda > 1$. The value of λ is then determined by a combination of the primary and residual secondary wall effects. However, the sheer brilliance of the Nikuradze data¹¹ is that it contained measurements over a broad enough range of Reynolds number values such that the six levels of roughness studied showed examples of the roughness being obscured by the thickness of the boundary layer at low Reynolds number values, on the one hand, and, on the other hand, punching through the boundary layer as the value of the Reynolds number increased and the boundary layer becomes more and more dissipated. Hence, the distinctive characteristic shape of the Θ vs Q_N curves in the QFFM, which are an alternative representation to that of the established Moody chart behavior.

In this limit, the QFFM equations reduce to the governing equations for flow in a circular pipe, and the dimensionless permeability Θ follows the characteristic curve of the pipe flow, including the distinct laminar-to-turbulent transition. This demonstrates how the QFFM naturally encompasses both packed beds and empty conduits within a single, unified theoretical framework.

V. CONCLUSION

This study has demonstrated the significant influence of particle roughness on pressure drop in randomly packed beds through comprehensive experimental validation and theoretical modeling using the Quinn fluid flow model (QFFM). The QFFM, which extends the classical Ergun equation by incorporating roughness effects, has been shown to accurately predict pressure drop across all flow regimes (creeping to turbulent) for both packed and empty conduits.

For polished spherical particles ($k = 0$), the QFFM achieved exceptional agreement with experimental data, with a mean absolute error (MAE) of approximately 1%. The wall normalization coefficient λ for these smooth particles was consistently close to unity ($\lambda \approx 1.003$), confirming negligible secondary wall effects. In contrast, rough cylindrical particles exhibited a measurable increase in λ to approximately 1.1, leading to a 20%–30% reduction in dimensionless permeability (Θ) compared to smooth particles. This reduction directly translates to higher pressure drops in rough-particle systems.

The analysis of irregularly shaped clinker particles revealed even more pronounced effects, with λ values exceeding 3–6 due to their high surface roughness ($k \sim 10^{-4}$ – 10^{-5} cm). These particles exhibited permeability reductions of 50%–70% relative to smooth-particle beds, underscoring the limitations of traditional models like the Ergun and Kozeny–Carman equations for irregular geometries. The QFFM's dimensionless analysis successfully unified these observations, bridging Nikuradze's pipe-flow data to closed conduits and differentiating between primary (viscous boundary layer) and secondary (roughness-induced) wall effects.

The QFFM solver's ability to back-calculate critical parameters (e.g., λ , ε_0) from limited experimental data and its robustness across diverse particle morphologies (spheres, cylinders, and clinker) highlight its potential for industrial applications involving rough or irregular particles. By quantifying roughness effects through λ and unifying predictions across flow regimes, the QFFM addresses a critical gap in

porous media flow modeling, offering a physics-based alternative to empirical correlations. Future work could explore extensions to poly-disperse systems and reactive flows, further broadening the model's applicability.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

V. Papkov: Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **H. M. Quinn:** Conceptualization (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **D. Pashchenko:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

NOMENCLATURE

Abbreviations

HQC	Hypothetical Q channel
MAE	Mean absolute error
QFFM	Quinn fluid flow model
RMSE	Root mean squared error

Greek symbols

γ_a	Area-weighted uniformity index
δ	Porosity normalization coefficient
Θ	Dimensionless permeability
ε	Q-Porosity function
ε_0	Conduit external volume fraction
ε_i	Conduit internal volume fraction
ε_p	Particle porosity
ε_t	Conduit total volume fraction
ε_{sk}	Conduit particle skeletal volume fraction
λ	Wall normalization coefficient
μ_s	Fluid superficial velocity
ν	Fluid absolute viscosity
ρ_f	Fluid density

Abbreviations

ΔP	Pressure loss
A	Viscous constant
A_h	Hydraulic cross-section area

B	Kinetic constant
C_Q	Fluid current for fluid flow in closed conduits
d_c	Hypothetical Q-channel diameter
d_p	Average spherical particle diameter equivalent
f_v	Viscous friction factor
G	Acceleration due to gravity
G^m	Fluid mass flow rate
I	Turbulent intensity
k_1	Universal viscous constant
k_2	Universal kinetic constant
L	Channel length
L_m	Turbulent length scale
n_k	Kinetic hydraulic force
n_p	Number of particles
n_v	Viscous hydraulic force
n_{pq}	Volume of the conduit expressed in terms of number of particles
P_Q	Normalized dimensionless pressure gradient
q	Volumetric flow rate
Q_N	Instantaneous fluid current
r_h	Fluid drag normalization coefficient
R_t	Turbulent viscosity ratio
Re_m	Modified Reynolds number

REFERENCES

- ¹D. Nemeec and J. Levec, "Flow through packed bed reactors: I. Single-phase flow," *Chem. Eng. Sci.* **60**, 6947–6957 (2005).
- ²A. G. Dixon, "Local transport and reaction rates in a fixed bed reactor tube: Endothermic steam methane reforming," *Chem. Eng. Sci.* **168**, 156–177 (2017).
- ³F. Janssen, M. B. Cardenas, A. H. Sawyer, T. Dammrich, J. Krietsch, and D. de Beer, "A comparative experimental and multiphysics computational fluid dynamics study of coupled surface-subsurface flow in bed forms," *Water Resour. Res.* **48**, 11982, <https://doi.org/10.1029/2012WR011982> (2012).
- ⁴F. Canovaro, E. Paris, and L. Solari, "Effects of macro-scale bed roughness geometry on flow resistance," *Water Resour. Res.* **43**, 5727, <https://doi.org/10.1029/2006WR005727> (2007).
- ⁵C. Voigt, E. Werzner, R. Fritsch, J. Hubálková, M. Hassanabadi, R. E. Aune, M. D. Innocentini, and C. G. Aneziris, "Influence of filter surface roughness on the pressure drop of ceramic foam filters," *Open Ceram.* **15**, 100379 (2023).
- ⁶R. Van Velden, S. Woudberg, E. Dumont, H. Fidler, and J. T. M. De Hosson, "The effect of particle surface roughness and biofilm accumulation on the pressure drop over a biofilter," *Chem. Eng. Sci.* **318**, 122185 (2025).
- ⁷S. Ergun, "Fluid flow through packed columns," *Chem. Eng. Prog.* **48**, 89 (1952).
- ⁸Y. Liu and X. Yu, "General formulation of drag force on assemblage of spherical particles in fluids: A critical review and a new empirical formula," *Phys. Fluids* **34**, 063307 (2022).
- ⁹A. Koekemoer and A. Luckos, "Effect of material type and particle size distribution on pressure drop in packed beds of large particles: Extending the Ergun equation," *Fuel* **158**, 232–238 (2015).
- ¹⁰N. Georgousis, J. Diriken, M. Speetjens, and C. Rindt, "Comprehensive review on packed-bed sensible heat storage systems," *J. Energy Storage* **121**, 116516 (2025).
- ¹¹J. Nikuradse, "Stromungsgesetze rauhen rohren," *VDI-Forschungsheft* **361**, 19 (1933).
- ¹²J. Kozeny, "Ueber kapillare leitung des wassers im boden," *Sitzungsber. Akad. Wiss. Wien* **136**, 271 (1927).
- ¹³P. C. Carman, "Fluid flow through granular beds," *Trans. Inst. Chem. Eng. London* **15**, 150–156 (1937).
- ¹⁴R. Schulz, N. Ray, S. Zech, A. Rupp, and P. Knabner, "Beyond Kozeny–Carman: Predicting the permeability in porous media," *Transp. Porous Media* **130**, 487–512 (2019).
- ¹⁵Z. Guo, Q. Fang, M. Nong, and X. Ren, "A novel Kozeny–Carman-based permeability model for hydrate-bearing sediments," *Energy* **234**, 121203 (2021).
- ¹⁶M. Wang, S. Bu, B. Zhou, B. Gong, Q. Lian, Z. Li, P. Zhang, D. Chen, and Q. Wang, "Effect of particle breakage on heat and mass transfer characteristics of packed bed at low Reynolds numbers: Pore-scale simulation," *Energy* **336**, 138402 (2025).
- ¹⁷M. Safari, R. Gholami, M. Jami, M. A. Ananthan, A. Rahimi, and W. S. Khur, "Developing a porosity-permeability relationship for ellipsoidal grains: A correction shape factor for Kozeny–Carman's equation," *J. Pet. Sci. Eng.* **205**, 108896 (2021).
- ¹⁸H. A. Nooruddin and M. E. Hossain, "Modified Kozeny–Carmen correlation for enhanced hydraulic flow unit characterization," *J. Pet. Sci. Eng.* **80**, 107–115 (2011).
- ¹⁹M. Rehman, M. B. Hafeez, and M. Krawczuk, "A comprehensive review: Applications of the Kozeny–Carman model in engineering with permeability dynamics," *Arch. Comput. Methods Eng.* **31**, 3843–3855 (2024).
- ²⁰L. Hua, H. Zhao, J. Li, J. Wang, and Q. Zhu, "Eulerian–Eulerian simulation of irregular particles in dense gas–solid fluidized beds," *Powder Technol.* **284**, 299–311 (2015).
- ²¹H. M. Quinn, "A reconciliation of packed column permeability data: Deconvoluting the Ergun papers," *J. Mater.* **2014**, 548482.
- ²²E. B. Hassan and J. Hoffmann, "Review on pressure drop through a randomly packed bed of crushed rocks," *Discov. Appl. Sci.* **6**, 126 (2024).
- ²³I. Macdonald, M. El-Sayed, K. Mow, and F. Dullien, "Flow through porous media—the Ergun equation revisited," *Ind. Eng. Chem. Fund.* **18**, 199–208 (1979).
- ²⁴N. Ahmed and D. K. Sunada, "Nonlinear flow in porous media," *J. Hydr. Div.* **95**, 1847–1858 (1969).
- ²⁵J. Prieur Du Plessis, "Analytical quantification of coefficients in the Ergun equation for fluid friction in a packed bed," *Transp. Porous Media* **16**, 189–207 (1994).
- ²⁶R. K. Niven, "Physical insight into the Ergun and Wen and Yu equations for fluid flow in packed and fluidised beds," *Chem. Eng. Sci.* **57**, 527–534 (2002).
- ²⁷C. Crawford and O. Plumb, "The influence of surface roughness on resistance to flow through packed beds," *J. Fluids Eng.* **108**, 343–347 (1986).
- ²⁸K. Allen, T. Von Backström, and D. Kröger, "Packed bed pressure drop dependence on particle shape, size distribution, packing arrangement and roughness," *Powder Technol.* **246**, 590–600 (2013).
- ²⁹T. Buchwald, G. Schmandra, L. Schützenmeister, T. Fraszczak, T. Mütze, and U. Peuker, "Gaseous flow through coarse granular beds: The role of specific surface area," *Powder Technol.* **366**, 821–831 (2020).
- ³⁰V. Papkov, N. Shadymov, and D. Pashchenko, "Gas flow through a packed bed with low tube-to-particle diameter ratio: Effect of pellet roughness," *Phys. Fluids* **36**, 027127 (2024).
- ³¹I. Karpilov and D. Pashchenko, "Steam methane reforming over a preheated packed bed: Heat and mass transfer in a transient process," *Therm. Sci. Eng. Prog.* **42**, 101868 (2023).
- ³²H. Quinn, "Quinn's law of fluid dynamics pressure-driven fluid flow through closed conduits," *Fluid Mech.* **5**, 39 (2019).
- ³³H. M. Quinn, "Quinn's law of fluid dynamics: Supplement #1—Nikuradze's inflection profile revisited," *Fluid Mech.* **6**, 1–14 (2020).
- ³⁴M. Nikitin, H. Quinn, and D. Pashchenko, "Flow through packed beds: Experimental and numerical validation of the Quinn fluid flow model," *Phys. Fluids* **37**, 084113 (2025).
- ³⁵H. M. Quinn, "Quinn's law of fluid dynamics: Supplement #2—Reinventing the Ergun equation," *Fluid Mech.* **6**, 15–29 (2020).
- ³⁶P. Forchheimer, "Wasserbewegung durch boden," *Z. Ver. Dtsch. Ing.* **45**, 1781–1788 (1901).
- ³⁷J. C. Giddings, *Dynamics of Chromatography: Part I: Principles and Theory* (Marcel Dekker, New York, 1965), Vol. 1.